

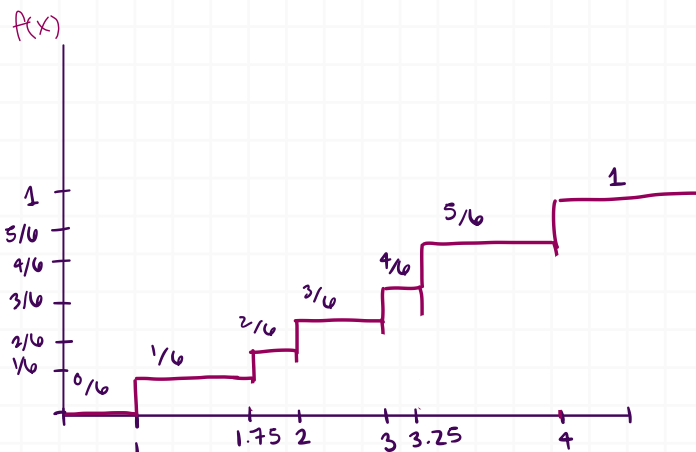
① [1, 1.75, 2, 3, 3.25, 4]

a) ECDF

$$\hat{F}(x) = \frac{\text{\# data pts} \leq x}{n} = \frac{\text{\# } n \leq x}{6}$$



$x < 1$	$F(x) = 0/6$
$x = 1$	$1/6$
$x = 1.75$	$2/6$
$x = 2$	$3/6$
$x = 3$	$4/6$
$x = 3.25$	$5/6$
$x = 4$	$6/6$



b) KDE for $h = 0.2$

$$\hat{f}(x) = \frac{\text{\# } n \text{ w/i } h \text{ of } x}{2 \cdot h \cdot n} \rightarrow \frac{1}{2nh} = \frac{1}{2 \cdot 6 \cdot 0.2} = \frac{1}{2.4} \approx 0.417$$

x_{pt}	$(x - 0.2 \text{ to } x + 0.2)$
1	0.8 - 1.2
1.75	1.55 - 1.95
2	1.8 - 2.2
3	2.8 - 3.2
3.25	3.05 - 3.45
4	3.8 - 4.2



c) Silverman $h = 1.84 \cdot SD \cdot n^{-0.2} \rightarrow (1.84)(1.107)(0.699) \approx 1.42$

$$\mu = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6} = \frac{13}{6} = 2.167$$

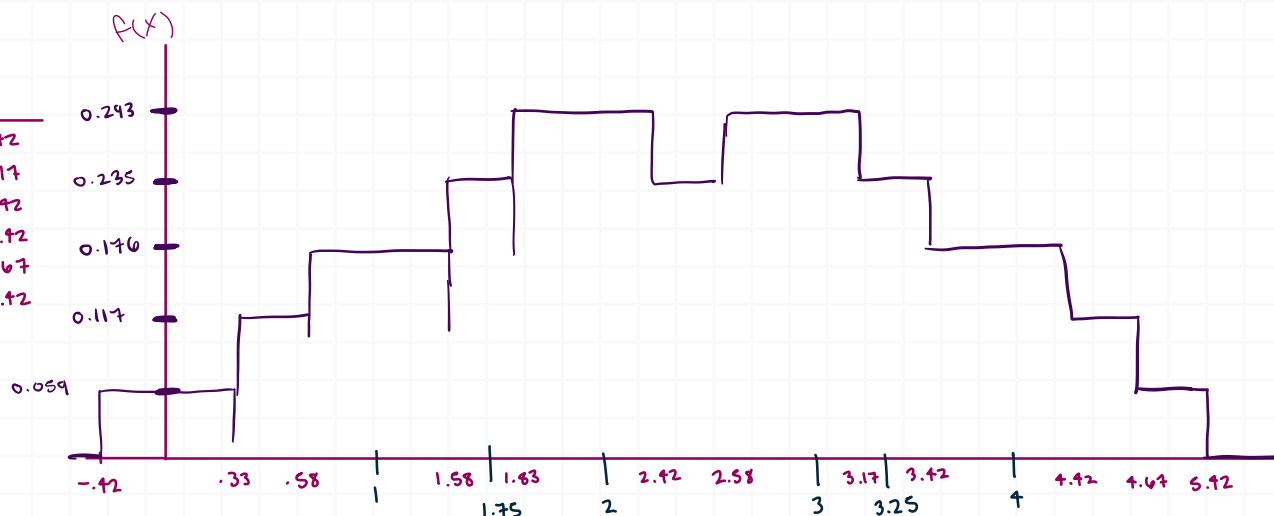
$$\sigma = \frac{(1 - 2.167)^2 + (1.75 - 2.167)^2 + (2 - 2.167)^2 + (3 - 2.167)^2 + (3.25 - 2.167)^2 + (4 - 2.167)^2}{6 - 1} \approx 1.107$$

$$h^{-0.2} \approx 0.699$$

$$\text{height} = \frac{1}{2 \cdot 6 \cdot 1.42} \approx 0.059$$

d) sketch c

x_{pt}	window
1	-0.42 - 2.42
1.75	.33 - 3.17
2	.58 - 3.42
3	1.58 - 4.42
3.25	1.83 - 4.67
4	2.58 - 5.42



e) b & d comparison

When h grows from 0.2 to 1.42 each box gets wider & shorter, 0.417 drops to 0.059, but the area of each box stays $1/6$ either way. They're both still uniform kernel estimates, so both are made of flat steps & have total areas of 1. When $h = 0.2$, the boxes are narrow & barely overlap, so you get tall spikes in the data with density of 0 in the gaps. This is an example of underfitting/smoothing b/c it implies certain values can never occur. With $h = 1.42$, the boxes overlap so much that they start & merge into the KDE we see above, peaking @ 2 & 3, and covering the whole range. (this could be seen as oversmoothed since it spreads probability below 0, but we don't know what an actually accurate model looks like so won't make claim).

② $[-1, \underbrace{-0.5, 0, 0.5}_{w/h}, 1]$

★ Supporting code for Q2 is in .ipynb file in same folder as PS PDF ★

a) $h=0.6$, $x=0$; $\hat{f}(h(x))$

$$\hat{f}(0) = \frac{\text{\# pt w/ } 0.6 \text{ of } 0}{2 \cdot 0.5 \cdot 0.6} = \frac{3}{6} = \boxed{0.5}$$

$$b) \mathbb{E}[\hat{f}(0)] = \frac{F(0+h) - F(0-h)}{2h} = \frac{F(0.6) - F(-0.6)}{1.2} \approx \boxed{0.376}$$

$$c) f(0) = \frac{1}{\sqrt{2\pi}} \approx \boxed{0.399} \quad (\mathbb{E}=0 \text{ into gaussian formula from exp/log notebook; "the bell curve"})$$

d) No, $\mathbb{E}[\hat{f}(0)] = 0.376 < 0.399 = f(0)$. The KDE is biased, underestimating $f(0)$ by 0.023 @ 0 on average

$$e) \mathbb{E}[\hat{f}(0)] = \frac{F(0.1) - F(-0.1)}{2(0.1)} \approx \boxed{0.398}$$

gap: $0.399 - 0.398 \approx \boxed{0.001} \rightarrow$ gap shrinks towards 0 as h gets smaller ($0.6 \rightarrow 0.023$ to $0.1 \rightarrow 0.001$)

f) $\hat{f}(h)$ is only consistent as h shrinks as n grows. As $h \rightarrow 0$, the bias gets smaller and smaller (shown in part e comparison). And as n grows, the randomness goes away. Silverman's rule shrinks h w/ n slowly enough to create this consistency. With a fixed h , the bias never fully goes away, so the KDE doesn't converge to $f(x)$ no matter how large n gets.

g) The main consideration is how much you can reduce bias without having too high of a variance (tradeoff). A smaller h lowers the bias, but if it's too small relative to sample size it will overfit & will be spiky & change a lot from sample to sample. A large h has a lower variance & higher bias. So, h should shrink as n grows. It is worth it to start w/ Silverman's rule & try a couple of values of h to see which estimate is most accurate.

③ $X = [5, 7, 7, 8, 10]$, $h = 1.5$

a) $X = 6, 6.5, 7, 8, 8.5$

$$\hat{f}(x) = \frac{1}{2nh} = \frac{1}{15}$$

X	window	(distance ≤ 1.5) pts inside	count	$\hat{f}(x)$
6	4.5 - 7.5	5, 7, 7	3	3/15
6.5	5 - 8	7, 7	2	2/15
7	5.5 - 8.5	7, 7, 8	3	3/15
8	6.5 - 9.5	7, 7, 8	3	3/15
8.5	7 - 10	8	1	1/15

b)
$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{X_i - x}{h}\right)^2\right)$$

$$\hat{f}(6) = 0.151$$

$$\hat{f}(6.5) = 0.169$$

$$\hat{f}(7) = 0.178$$

$$\hat{f}(8) = 0.167$$

$$\hat{f}(8.5) = 0.151$$

★ supporting code in .ipynb file ★

c) The biggest jump in part a was from $x = 8.0$ to $x = 8.5$, where it drops from 0.2 to 0.067. Over the same stretch, the Gaussian only moves from 0.167 to 0.151. The uniform kernel is a flat box with vertical edges, so a point either contributes its full height, or nothing. From $x = 8$ to 8.5 , the two 7s go from a distance of 1 to a distance of exactly 1.5, which is right on the box edge so they both are excluded. This is what creates the drastic jump. The Gauss kernel is a bell curve with no edges, so every point contributes something and fades gradually as the distance grows. This creates a gradual decline instead of sudden jumps in this case, which gives a smoother KDE.

d) Uniform KDEs are easier to compute by hand, and they're simple/transparent, so easily explainable. It's directly related to the ECDF & each pt only affects a limited range. Gaussian is smoother, and doesn't produce artificial jumps or sudden changes that aren't real features of the data. For these reasons, the Gauss is preferred for plotting/presenting/interpreting a density at any sample size.

★ Q4 IN NOTEBOOK ★